SCAC: A Benchmark Towards Robust Semantic Correspondence

Supplementary Material

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A Dataset Construction Details

Building upon the clean subset, we further augment it with four key dimensions of adverse conditions: geometric variations, blur and noise, digital artifacts, and environmental changes. (1) **Geometry.** It refers to transformations that modify the spatial layout and intrinsic structure of the object within an image, including Rotation, Viewpoint, and Scale. These transformations are performed following the code provided by [5]. For Rotation, Each image is randomly rotated by an angle between 0° and 360°. For Viewpoint, Images undergo random viewpoint transformations by selecting source and target points within a 100–300 pixel padding range to simulate diverse viewing angles. For Scale, Images are randomly scaled by a factor between 0.1 and 0.9 using a transformation matrix. All transformed images are then composited onto a background to ensure consistency.(2) **Blur&Noise.** This category involves degradations that disrupt fine-grained structural cues, comprising Defocus blur, Motion blur, and Gaussian noise. These perturbations are generated using the publicly available implementation provided by [3]. Specifically, Defocus Blur is applied using a circular kernel smoothed with Gaussian blur to create a uniform defocus effect; Motion Blur is simulated by shifting pixels along a given angle with weighted accumulation to mimic motion trails; Gaussian Noise is added by overlaying Gaussian-distributed noise on a normalized image, simulating sensor-level interference. (3) **Digital.** These distortions address artifacts that degrade visual fidelity and pixel-level consistency arising from digital image processing pipelines, including JPEG Compression, Pixelation, and Contrast. They are implemented using the publicly available code from [3]. In particular, JPEG Compression is simulated by converting the image to RGB (if needed), saving it at low quality, and reloading it to introduce compression noise; Pixelation is done by resizing the image to a lower resolution and then scaling it back using nearest-neighbor interpolation, resulting in a blocky visual effect; Contrast is adjusted by normalizing the image, computing the mean, and applying a small scaling factor to pixel deviations, followed by clipping and rescaling, thereby simulating contrast degradation.(4) **Environment.** This group encompasses adverse conditions caused by dynamic environmental factors that alter lighting, scattering, or occlusion in images, consisting of Rain, Fog, Snow, Dark and Strong sunlight. These degradations are generated using the open-source implementation from [1–4]. In detail, Rain is produced by blending resized, Gaussian-blurred rain streak images with the original image weighted by rain intensity; Fog is synthesized using a depth-guided haze model based on estimated depth maps and parameters including transmission and atmospheric light; Snow is simulated by overlaying resized snow mask images through weighted combination; Dark scenes are created by applying gamma correction with a low gamma value to decrease overall brightness; and Strong Sunlight is modeled by increasing brightness in the HSV color space for color images or adding an offset for grayscale images, followed by clipping and scaling back to the original range.

B Experiments Details

B.1 Experimental Set-up

Robustness Evaluation: All semantic correspondence methods are implemented using their official GitHub repositories with default pretrained models. And all experiments are conducted on an NVIDIA GeForce RTX 3090 GPU to ensure consistency in runtime performance. To ensure the validity of our implementation, we first reproduce baseline results on the original SPair-71K dataset, confirming consistency with the performance reported in the respective original papers. We then evaluate all methods on SCAS dataset proposed in this paper. To ensure fair comparisons, we use all methods without any hyperparameter tuning or architectural modifications, evaluating them under identical testing conditions.

Robustness Enhancement: To investigate robustness enhancement, we adopt an experimental setup as [8] and employ the Stable Diffusion v1-5 model as the feature extractor. The default number of time steps for the model is set to $t = 100$, and it receives inputs at a resolution of 960×960. The model is optimized using AdamW [6] with a weight decay of 0.001. We adopt the one-cycle learning rate scheduler [7], with a peak learning rate of 1.25×10^{-3} and 30% of the cycle dedicated to the warm-up phase. All experiments are conducted on a single NVIDIA RTX 3090 GPU.

C Additional Analyses

C.1 Accuracy of Semantic Correspondence Methods

Figure 1 presents the performance of various semantic correspondence methods across four scenarios: Geometry, Blur&Noise, Digital, and Environment. Across all methods, a consistent performance decline is observed when moving from clean to adverse conditions within each scenario. The decline is most pronounced in the Geometry scenario, particularly under the rotation setting, where performance has decreased by an average of 40%. Similarly, the viewpoint setting exhibits a notable drop, with an average decrease of 28.20%. Additionally, significant performance reductions are observed in the gaussian noise scenario (average decline of 16.81%) and in the haze scenario (average decline of 17.22%). In contrast,

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the decline is less pronounced in the Environment scenario, where methods exhibit minimal performance degradation in snow, rain, and strong sunlight settings. Specifically, accuracy drops by only 1.59%, 0.08%, and 1.37% respectively, indicating that the methods maintain relatively stable performance under these conditions.

C.2 Absolute and Relative Robustness Analysis

In most scenarios, the LVM-based fine-tuning method achieves higher absolute robustness than other approaches, particularly in the scale setting of the Geometry scenario (Figure 2) and the defocus blur setting of the Blur & Noise scenario (Figure 3). In contrast, the LVM-based unsupervised method generally exhibits greater relative robustness. Exceptions are limited to a few settings, such as contrast setting in the Digital scenario (Figure 4) and haze setting in the Environment scenario (Figure 5), where the fine-tuning method outperforms in both absolute and relative robustness. In contrast, the supervised approach consistently performs worse in both absolute and relative robustness compared to the other methods.

C.3 Robustness of DINO vs Stable Diffusion

Methods incorporating Stable Diffusion exhibit superior absolute robustness across fourteen challenging conditions. However, in scenarios involving geometric transformations (Table 1), particularly rotation, DINOv2 achieves the highest accuracy, outperforming other approaches. Furthermore, the DINO series demonstrates greater relative robustness across these conditions. However, under degradation scenarios such as Blur & Noise (Table 2), especially defocus blur and gaussian noise, DIFT achieves the best performance. Meanwhile, under adverse environmental conditions (Table 4) including snow, rain, and strong sunlight, methods leveraging Stable Diffusion consistently outperform other methods.

References

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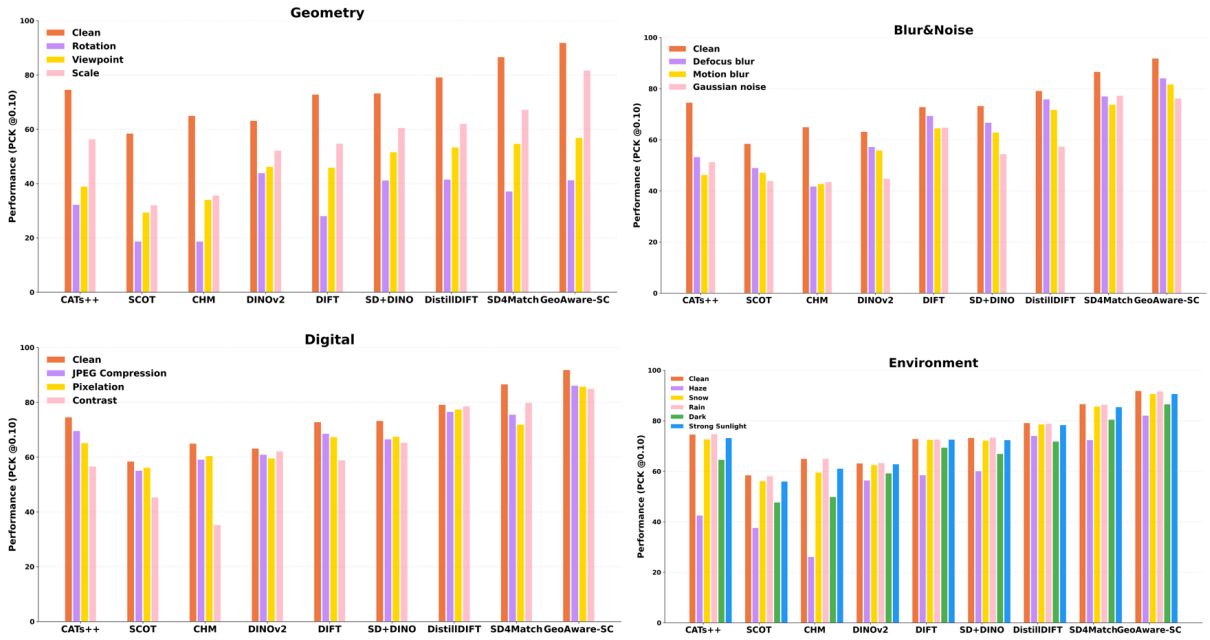

Figure 1: Accuracy of various semantic association methods across four scenarios. The figure consists of four bar charts, each representing the performance of different methods in a specific scenario.

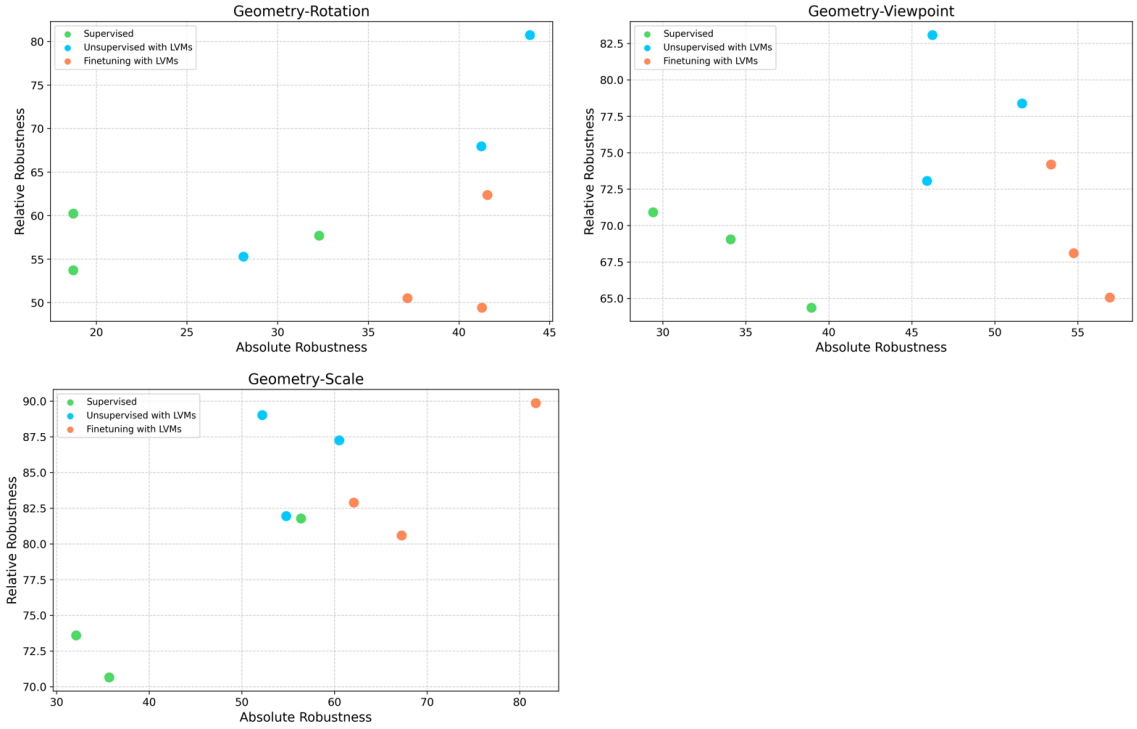

Figure 2: The scatter plot demonstrates the relative and absolute robustness of three categories of methods when subjected to geometric transformations, including rotation, viewpoint changes, and scale variations.

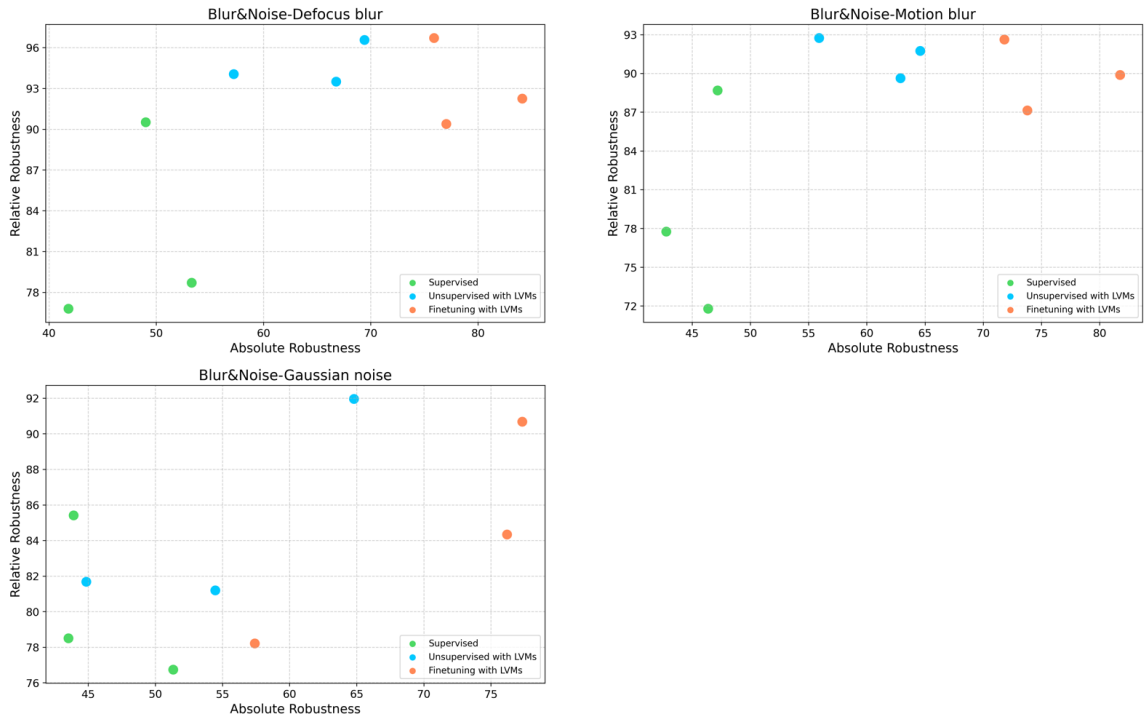

Figure 3: The scatter plot demonstrates the relative and absolute robustness of three categories of methods when subjected to blur and noise transformations, including defocus blur, motion blur, and gaussian noise.

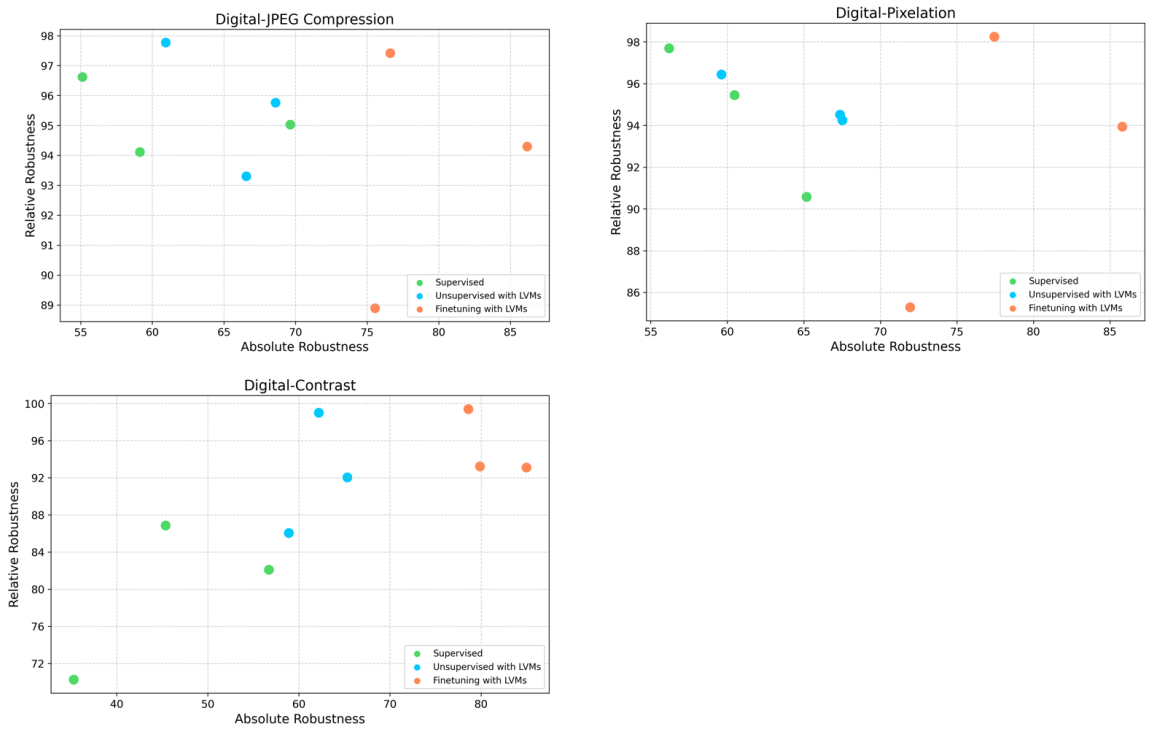

Figure 4: The scatter plot demonstrates the relative and absolute robustness of three categories of methods when subjected to digital transformations, including JPEG compression, pixelation, and contrast changes.

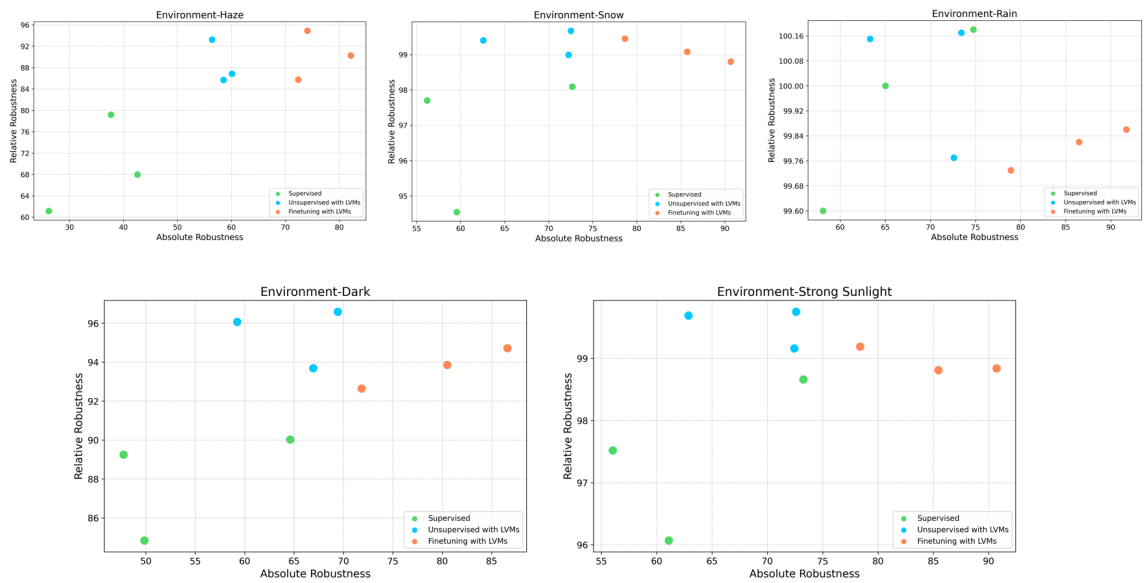

Figure 5: The scatter plot demonstrates the relative and absolute robustness of three categories of methods when subjected to environmental transformations, including haze, snow, rain, darkness, and strong sunlight.

Category	Method	Clean	Absolute Robustness				Relative Robustness			
			Rotation	Viewpoint	Scale	Avg	Rotation	Viewpoint	Scale	Avg
D	DINOv1	46.10	18.72	25.62	28.80	24.38	72.62	79.52	82.70	78.28
	DINOv2	63.17	**43.91**	46.24	52.19	47.45	**80.74**	**83.07**	**89.02**	**84.28**
G	SDv1.5	68.71	28.76	42.80	51.55	41.04	60.05	74.09	82.84	72.33
	DIFT(SDv2.1)	72.84	28.11	45.91	54.79	42.94	55.27	73.07	81.95	70.10
D&G	SD+DINO	**73.26**	41.23	**51.64**	**60.51**	**51.13**	67.97	78.38	87.25	77.87

Table 1: Comparison of absolute and relative robustness for DINO, Stable Diffusion, and Stable Diffusion+DINO methods under geometric transformations, including rotation, viewpoint, and scale.

Category	Method	Clean	Absolute Robustness				Relative Robustness			
			Defocus blur	Motion blur	Gaussian noise	Avg	Defocus blur	Motion blur	Gaussian noise	Avg
D	DINOv1	46.10	38.32	36.06	37.20	37.19	92.22	89.96	91.10	91.09
	DINOv2	63.17	57.23	55.91	44.85	52.66	94.06	**92.74**	81.68	89.49
G	SDv1.5	68.71	57.23	49.18	48.25	51.55	88.52	80.47	79.54	82.84
	DIFT(SDv2.1)	72.84	**69.41**	**64.59**	**64.80**	**66.27**	**96.57**	91.75	**91.96**	**93.43**
D&G	SD+DINO	**73.26**	66.76	62.89	54.46	61.37	93.50	89.63	81.20	88.11

Table 2: Comparison of absolute and relative robustness for DINO, Stable Diffusion, and Stable Diffusion+DINO methods under Blur&Noise conditions, including defocus blur, motion blur, and Gaussian noise.

Category	Method	Clean	Absolute Robustness				Relative Robustness			
			JPEG Compression	Pixelation	Contrast	Avg	JPEG Compression	Pixelation	Contrast	Avg
D	**DINOv1**	46.10	43.03	43.71	34.60	40.45	96.93	**97.61**	88.50	94.35
	DINOv2	63.17	60.94	59.61	62.18	60.91	**97.77**	96.44	**99.01**	**97.74**
G	**SDv1.5**	68.71	53.47	57.46	38.92	49.95	84.76	88.75	70.21	81.24
	DIFT(SDv2.1)	72.84	**68.60**	67.36	58.89	64.95	95.76	94.52	86.05	92.11
D&G	**SD+DINO**	**73.26**	66.56	**67.51**	**65.31**	**66.46**	93.30	94.25	92.05	93.20

Table 3: Comparison of absolute and relative robustness for DINO, Stable Diffusion, and Stable Diffusion+DINO methods under digital conditions, including JPEG compression, pixelation, and contrast.

Category	Method	Clean	Absolute Robustness						Relative Robustness					
			Haze	Snow	Rain	Dark	Strong Sunlight	Avg	Haze	Snow	Rain	Dark	Strong Sunlight	Avg
D	**DINOv1**	46.10	25.57	43.45	45.07	37.63	43.92	39.13	79.47	97.35	98.97	79.47	97.82	93.03
	DINOv2	63.17	56.42	62.57	62.18	59.24	62.86	60.65	**93.25**	99.40	99.01	**93.25**	99.69	**97.48**
G	**SDv1.5**	68.71	42.37	67.07	67.93	61.31	67.40	61.43	73.66	98.36	99.22	73.66	98.69	92.60
	DIFT(SDv2.1)	72.84	58.54	**72.51**	72.96	**69.43**	**72.59**	**69.21**	85.70	**99.67**	**100.12**	85.70	**99.75**	96.37
D&G	**SD+DINO**	**73.26**	**60.11**	72.25	**73.11**	66.95	72.42	68.97	86.85	98.99	99.85	86.85	99.16	95.71

Table 4: Comparison of absolute and relative robustness for DINO, Stable Diffusion, and Stable Diffusion+DINO methods under environmental conditions, including haze, snow, rain, dark, and strong sunlight.